



IT Employee Lifecycle Automation System

*Automate Onboarding & Offboarding — Zero Manual Steps, Zero Orphaned
Accounts*

Author: Stephen Torrijas

Project: IT Employee Lifecycle Automation System

The Problem

IT administrators spend hours **manually creating Active Directory accounts** for new hires — often *after* the employee's first day — and frequently miss offboarding notifications, leaving former employees with active credentials and access to sensitive resources. Manual provisioning leads to inconsistent permissions, audit failures, and significant security risk.

The Solution

The **IT Employee Lifecycle Automation System** continuously monitors OrangeHRM for personnel changes and automatically executes corresponding Active Directory actions. New hires get accounts on **Day 1**. Departing employees are disabled within **minutes of termination**. Every action is logged, auditable, and reviewable through a clean web dashboard.

Key Features

1 OrangeHRM Sync

Polls OrangeHRM at a configurable interval, syncing employee name, department, and job title in real time.

2 Auto AD Provisioning

Creates Active Directory accounts via LDAP with correct OU placement, display name, email, and enabled status.

3 Role-Based Group Assignment

Maps HR department + job title combinations to AD security groups; assignments applied automatically on creation.

4 Auto Deprovisioning

Detects terminations and disables AD accounts immediately, removing group memberships to eliminate orphaned access.

5 Task Management

Tracks every provisioning task with status (Pending → In Progress → Completed/Failed), retry logic, and priority flagging.

6 Audit Log + CSV Export

Full chronological record of all lifecycle events with date-range filtering and one-click CSV export for compliance reviews.

7 Notification System

8 Access Request Workflow

Real-time in-app notifications with unread badge counter; notifications toggle-able via system settings.

Department managers submit additional-access requests; IT Admins approve or reject with full audit trail.

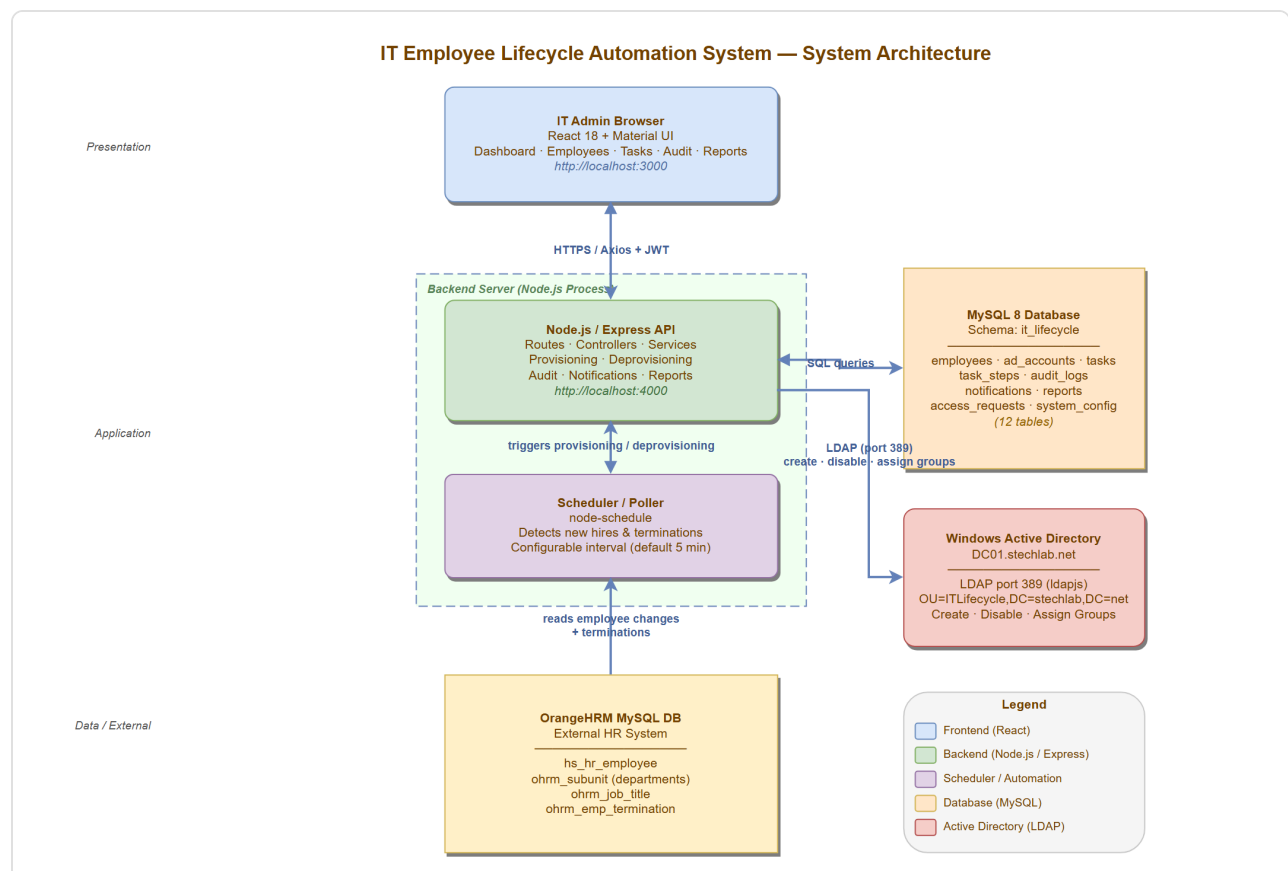
9 Compliance Reports

Generate activity summary reports filtered by date range and department; download as CSV.

10 System Configuration UI

Adjust polling interval, retry limit, and notification settings at runtime — no code changes required.

System Architecture



Screen Snapshots

Dashboard — Real-time Task Statistics & OrangeHRM Sync Status

IT Lifecycle System

5

Dashboard

Employees

Tasks

Mappings

Audit Log

Reports

Access Requests

5 Notifications

Settings

Dashboard

3

Total Employees

0

Pending Tasks

4

Completed Today

2

Failed Tasks

OrangeHRM Sync Status

Sync Now

● OrangeHRM DB: Connected

Employees in OrangeHRM: 3

Last sync: 3/8/2026, 3:00:00 PM

Departments synced: 1

SUCCESS

Employees in OrangeHRM

Emp #	First Name	Last Name	Email	Job Title	Department	Status
1	Stephen	Torrijas	sttorrij@lu.com	IT	—	ACTIVE
2	John	Doe	—	Accountant	Accounting	ACTIVE
3	Mary	Sue	—	HR Assistant	—	ACTIVE

Recent Tasks

#9 — Mary Sue | ONBOARDING

COMPLETED

#8 — Stephen Torrijas | ONBOARDING

COMPLETED

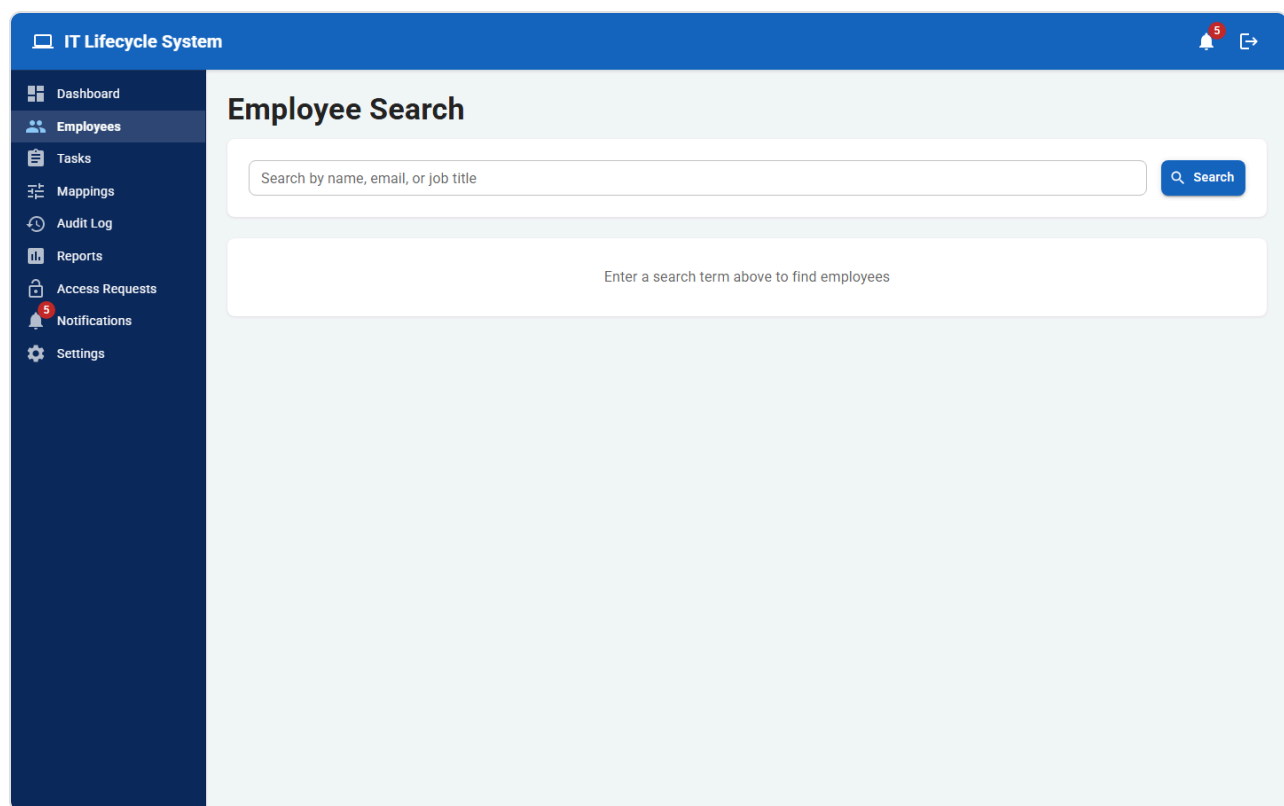
#7 — John Doe | ONBOARDING

COMPLETED

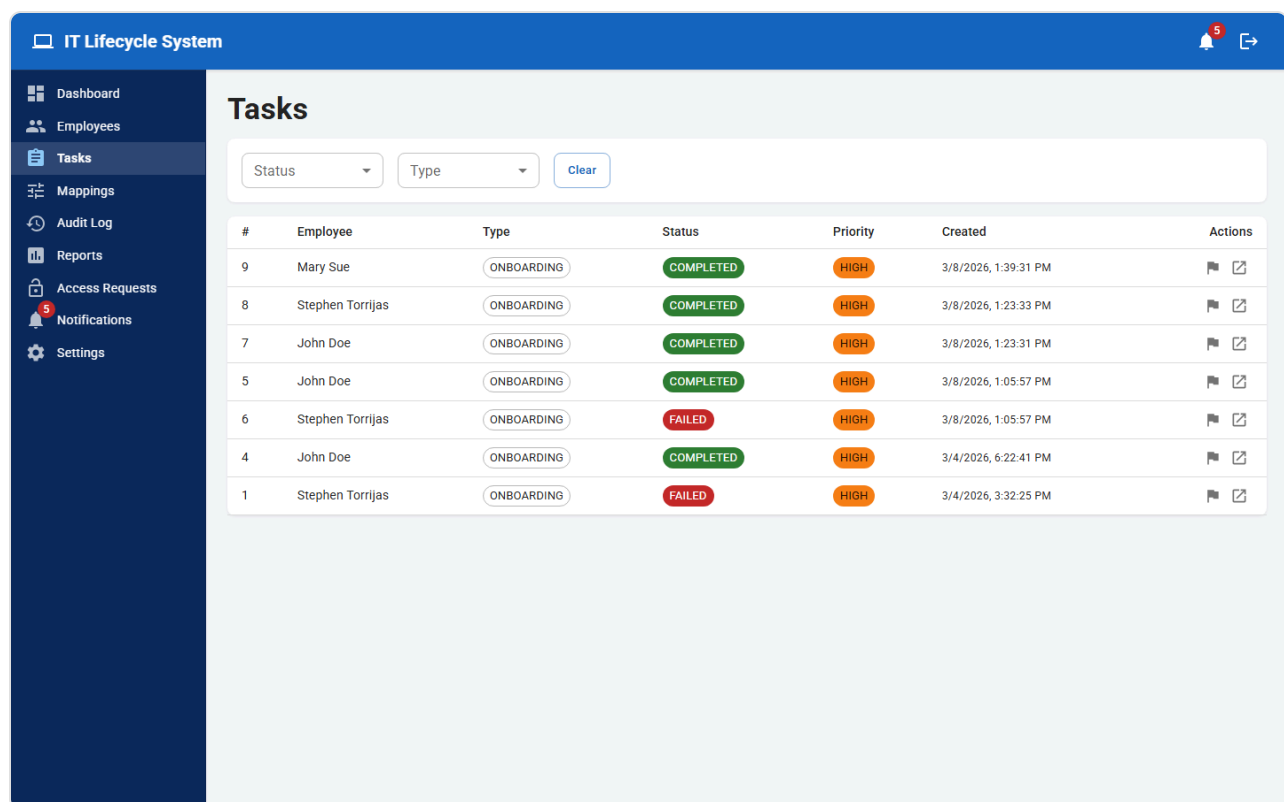
#5 — John Doe | ONBOARDING

COMPLETED

Employee Search — Clickable Rows with AD Account Detail Modal



Task List — Status/Type Filters, Priority Badges, Flag-Critical Button



Audit Log — Date Range Filtering & CSV Export

IT Lifecycle System

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Settings

Audit Log

Filter by actor

Filter by action

From date
mm/dd/yyyy

To date
mm/dd/yyyy

Apply Filters

Timestamp	Actor	Action	Target	Detail
3/8/2026, 1:39:31 PM	system	EMPLOYEE_PROVISIONED	marysue	Employee Mary Sue (3) provisioned
3/8/2026, 1:23:33 PM	system	EMPLOYEE_PROVISIONED	stephentorrijas	Employee Stephen Torrijas (1) provisioned
3/8/2026, 1:23:31 PM	system	EMPLOYEE_PROVISIONED	johndoe	Employee John Doe (2) provisioned
3/8/2026, 1:06:01 PM	system	PROVISIONING_ERROR	1	createUser failed: Unwilling To Perform
3/8/2026, 1:05:57 PM	system	EMPLOYEE_PROVISIONED	johndoe	Employee John Doe (2) provisioned
3/4/2026, 6:22:41 PM	system	EMPLOYEE_PROVISIONED	johndoe	Employee John Doe (2) provisioned
3/4/2026, 6:19:22 PM	system	PROVISIONING_ERROR	2	createUser failed: Invalid Attribute Syntax
3/4/2026, 4:40:04 PM	system	PROVISIONING_ERROR	2	ldap://<ad-host> is an invalid LDAP url (scope)
3/4/2026, 3:32:29 PM	system	PROVISIONING_ERROR	1	name (string) required

1-9 of 9

System Requirements

Component	Requirement
Runtime	Node.js 18+ (tested on v25.7.0)
Database	MySQL 8.0+ (schema: it_lifecycle)
HR System	OrangeHRM 5.x (direct DB access)
Directory	Windows Server Active Directory (LDAP port 389)
OS	Windows Server 2019/2022 (for AD integration)
Browser	Chrome 120+, Edge 120+, Firefox 120+

Feature-to-Requirement Mapping

Feature	Requirement(s)
OrangeHRM Sync + Auto Provisioning	REQ-1, REQ-2, NFR-1
Role-Based Group Assignment	REQ-3, REQ-6
Auto Deprovisioning on Termination	REQ-4, REQ-5
Task Management + Retry Logic	REQ-7, NFR-5, NFR-6

Feature	Requirement(s)
Audit Log with CSV Export	REQ-9, NFR-9
Notification System	REQ-15, NFR-3
Access Request Workflow	REQ-14, REQ-7
Compliance Report Generation	REQ-10
System Configuration UI	REQ-8, NFR-10
Employee Search with AD Detail	REQ-11, REQ-13